

# Exact Flat-Band Stiffness from Pair Mobility in a QGN Model

Finite-size obstructions, a many-body reduction theorem, and a proof of the Gao-Han-Khalaf formula for Model II

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## Abstract

Gao, Han, and Khalaf conjectured that the thermodynamic flat-connection stiffness of a quantum-geometric-nesting (QGN) superconductor is exactly the hard-core filling factor times the inverse two-particle pair mass. We prove this equality for their projected-Hubbard Model II and derive a broader restricted reduction theorem. The route to the theorem passes through two exact boundary cases: a period-circumference winding deformation shows that finite-torus twist curvature is underdetermined unless a displacement lift is specified, while a fixed-local reducible QGN model shows that locality and positive pair mass do not suffice without irreducibility of the paired ground branch. Under H1-H4 in the Hermitian positive-square class, the fixed-pair-number Hessian obeys  $C^{(n)} = \frac{n(L-n)}{L-1}C^{(1)} + (n^2 - \frac{n(L-n)}{L-1})\Gamma$ , where the longitudinal tensor  $\Gamma$  is positive semidefinite and is a pure winding observable under a translation-covariant flat connection. In the projected PSD-Hubbard subclass, connectedness of the projected-orbital frame is equivalent to uniqueness of the antisymmetrized-geminal-power ground state in every nonsingular filling sector. For Model II, the frame is connected on every rectangular torus for  $0 < \xi < \pi/2$ ,  $\Gamma = 0$  exactly with no finite-size correction, and the lifted finite-volume inverse mass equals the thermodynamic pair mass,  $(m_{\text{pair}}^{-1})_{ij} = |U|\xi^2\delta_{ij}/8$ , for every  $N_x, N_y \geq 3$ . Hence  $\kappa_{ij}^{(n)} = [V/(V-1)]\nu(1-\nu)|U|\xi^2\delta_{ij}/8$  at finite size and  $D_{s,ij} = \nu(1-\nu)(m_{\text{pair}}^{-1})_{ij}$  in the thermodynamic flat-connection definition used by Gao, Han, and Khalaf.

## 1. Introduction and headline theorem

Gao, Han, and Khalaf used the reduced-density-matrix bootstrap to obtain rigorous lower bounds on the zero-temperature flat-connection response of frustration-free flat-band superconductors. In every QGN example they tested, the lower bound saturated a variational upper bound determined by the two-particle pair mass. They consequently conjectured

$$D_{s,ij} = \frac{N_{\text{flat}}}{2} \nu(1-\nu) (m_{\text{pair}}^{-1})_{ij}, \quad (1)$$

and explicitly left a rigorous proof to future work [1]. The content of (1) is unusually strong: it says that a finite-density many-body response is fixed by one bound pair, with no dynamical dressing beyond the hard-core filling factor.

This paper gives a complete proof for the projected positive-semidefinite Hubbard realization of their Model II and, at the same time, identifies the exact hypotheses that make the reduction work.

**Headline theorem.** Let

$$H_{\text{II}}(k) = -t[\sigma_x \sin \alpha_k + \sigma_y \cos \alpha_k], \quad \alpha_k = \xi(\cos k_x + \cos k_y), \quad (\text{T1})$$

and project the positive-semidefinite Hubbard interaction onto its lower time-reversal-related flat bands. For  $0 < \xi < \pi/2$ , every rectangular torus with  $N_x, N_y \geq 3$ ,  $V = N_x N_y$ , and  $1 \leq n \leq V-1$  has a unique analytic AGP ground branch with exact curvature

$$\kappa_{ij}^{(n)}(V) = \frac{V}{V-1} \nu(1-\nu) \frac{|U|\xi^2}{8} \delta_{ij}, \quad \nu = \frac{n}{V}. \quad (\text{T2})$$

For  $\mathbf{N} = (N_x, N_y)$ , the lifted finite-volume one-pair Hessian defined in Section 10 is independent of  $\mathbf{N}$  and equals the thermodynamic pair mass:

$$(m_{\text{pair}, \mathbf{N}}^{-1})_{ij} = (m_{\text{pair}}^{-1})_{ij} = \frac{|U|\xi^2}{8} \delta_{ij} > 0. \quad (\text{T3})$$

so along every regular sequence with  $n_V/V \rightarrow \nu \in (0, 1)$ ,

$$D_{s,ij} = \nu(1-\nu)(m_{\text{pair}}^{-1})_{ij}. \quad (\text{T4})$$

Here  $D_s$  is exactly the thermodynamic ground-energy curvature under the flat gauge connection used in Ref. [1]. The finite-size factor  $V/(V-1)$  is explicit and universal.

The proof emerged through the following sequence.

| Stage                         | Exact conclusion                                                        | Role in the final theorem                                                 |
|-------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Winding counterexample        | Untwisted finite-torus samples do not determine twist curvature.        | The displacement-resolved Peierls lift must be part of the model.         |
| Reducible QGN counterexample  | A fixed-local QGN model can have zero stiffness and positive pair mass. | Connectivity or branch uniqueness is essential.                           |
| Connected-model search        | All surviving deviations reduce to one nonnegative longitudinal tensor. | Suggests an exact algebraic reduction rather than another counterexample. |
| General Hermitian-QGN theorem | $C^{(n)} = \rho_n C^{(1)} + (n^2 - \rho_n) \Gamma$ .                    | Isolates the only finite-size defect.                                     |
| H4 connectivity theorem       | Connected projected frame $\iff$ unique AGP zero mode.                  | Removes the branch hypothesis in projected Hubbard models.                |
| Model-II dispersion lemma     | Pair mass is exact and positive; $\Gamma = 0$ exactly.                  | Closes every remaining hypothesis and proves (T2)-(T4).                   |

The standard QGN ideal Hamiltonians use Hermitian local null operators and a positive interaction kernel [2]. The abstract theorem below is restricted to that positive-square class, a full-rank equal-singular-value pairing matrix, and a specified translation-covariant twist lift. The final Model-II result is unconditional in the range stated above.

**Notation.** Throughout,  $L$  is the number of collective pair orbitals,  $\mathbf{N} = (N_1, \dots, N_d)$  are torus side lengths, and  $V = \prod_j N_j$  is the number of unit cells. The one-spin flat-band space is denoted  $\mathcal{V}$ . The filling is always  $\nu = n/L$ ; for Model II,  $L = V$ . In Section 2.1,  $g > 0$  denotes the magnitude of an attractive coupling. From Section 2.2 onward,  $U > 0$  denotes the coefficient of a positive-semidefinite square, while the Model-II formulas use  $|U|$ .

## 2. Two exact obstructions and the corrected theorem class

The two counterexamples are not discarded preliminaries. Together they identify the two independent ways in which an overbroad finite-size theorem can fail: the twist may be underdetermined, or the paired ground branch may be reducible.

## 2.1 Winding/lift obstruction

On an  $M$ -point one-dimensional torus, consider the smooth rank-one spinor

$$u_{\uparrow,c}^{(M,q)}(k) = \begin{pmatrix} \cos[\pi/4 + c \sin(qMk)] \\ e^{i\phi(k)} \sin[\pi/4 + c \sin(qMk)] \end{pmatrix}, \quad u_{\downarrow}(k) = u_{\uparrow}^*(-k). \quad (\text{O1})$$

At every allowed momentum  $k_m = 2\pi m/M$ , the deformation is invisible because  $\sin(qMk_m) = 0$ . Under the continuous shift  $k_m \mapsto k_m + A$ , however,

$$\sin[qM(k_m + A)] = \sin(qMA) \quad (\text{O2})$$

is independent of  $m$ . For a projected onsite attraction of magnitude  $g > 0$ , the shifted deformation angle  $x_A = c \sin(qMA)$  is independent of  $k_m$ . The two orbital pair amplitudes obey  $\cos(\pi/4 + x_A) \cos(\pi/4 - x_A) = \sin(\pi/4 + x_A) \sin(\pi/4 - x_A) = \frac{1}{2} \cos(2x_A)$ . Every projected interaction matrix element therefore acquires the same squared factor, giving the exact operator factorization

$$H_c(A) = \cos^2[2c \sin(qMA)] H_0(A). \quad (\text{O3})$$

If  $E_n(0) = -gn/2$ , then

$$E_{n,c}''(0) = E_{n,0}''(0) + 4gc^2q^2M^2n. \quad (\text{O4})$$

Consequently, with  $\rho_n = n(M - n)/(M - 1)$ ,

$$\boxed{E_{n,c}'' - \rho_n E_{1,c}'' = (E_{n,0}'' - \rho_n E_{1,0}'') + 4gc^2q^2M^2 \frac{n(n-1)}{M-1}.} \quad (\text{O5})$$

The explicit four-cell certificate uses  $M = 4$ ,  $q = 1$ ,  $c = 1/8$ ,  $\phi(k) = (\pi/2) \sin k$ , and  $n = 2$ . Equation (O5) then gives a raw extra curvature  $2g/3$ ; with the finite-volume convention  $\kappa = E''/(4M)$ , the exact normalized defect is  $g/24$ . The branch is unique and gapped. A nonresonant  $M = 8$ , harmonic-four control obeys the filling law, whereas the resonant harmonic-eight deformation violates it again. The failure is therefore not caused by a generic high harmonic; it is a period-circumference winding resonance.

The structural conclusion is more important than the example:

The untwisted finite-torus Hamiltonian fixes Fourier alias sums, not the displacement moments needed for a continuous twist. A finite-size curvature theorem must specify the displacement lift or the complete family  $P_A$ .

This example is not a fixed-local counterexample along a thermodynamic sequence because its resonant displacement grows with  $M$ . Fixed finite range removes the resonance once the torus is sufficiently large, and uniform exponential locality suppresses it exponentially.

## 2.2 Connectivity/reducibility obstruction

A distinct obstruction survives fixed locality. Take

$$u_{\uparrow}(k) = \begin{pmatrix} \cos k \\ \sin k \end{pmatrix}, \quad u_{\downarrow}(k) = u_{\uparrow}^*(-k), \quad (\text{O6})$$

whose projector has range two and obeys  $P(k + \pi) = P(k)$ . On every even torus, the band space decomposes into even and odd cell components. The twist-stable UPC holds exactly, and the interaction is the projected positive-semidefinite Hubbard Hamiltonian

$$H(A) = \frac{U}{2} \sum_{R,\alpha} [\bar{n}_{R\alpha\uparrow}(A) - \bar{n}_{R\alpha\downarrow}(A)]^2. \quad (\text{O7})$$

At half filling, the state with one component completely paired and the other empty is annihilated by every square factor for every  $A$ . Therefore

$$E_0(A) = 0, \quad D_s(M, \nu = 1/2) = 0. \quad (\text{O8})$$

The analytic one-pair branch through  $Q = 0$  is nevertheless

$$E_{\text{pair}}(Q) = \frac{U}{2} \sin^2 Q, \quad m_{\text{pair}}^{-1} = U. \quad (\text{O9})$$

Only the local curvature of this branch at  $Q = 0$  is used; no assertion that it remains the global lowest eigenvalue for all  $Q$  is needed. Thus

$$\boxed{R_M(1/2) = 0} \quad (\text{O10})$$

for every even  $M \geq 6$ . This is a fixed, strictly local, thermodynamically persistent discrepancy, but it occurs because the ground space is reducible and degenerate. On odd tori, where the step-two graph is connected, the reduction law returns.

The corrected theorem class must therefore include both:

1. a specified fixed-local twist lift; and
2. an irreducible, uniquely continued AGP ground branch.

The remainder of the paper proves that, in the standard Hermitian positive-square class, these repairs remove both known failure mechanisms and leave one explicitly controlled longitudinal winding tensor.

### 3. Finite-dimensional QGN setup

Let  $\mathfrak{h}$  be a one-particle Hilbert space of dimension  $2L$ . Let

$$J^T = -J, \quad J^\dagger J = I_{2L} \quad (2)$$

be a skew-unitary pairing matrix. Define

$$\eta^+ = \frac{1}{2} \sum_{a,b=1}^{2L} J_{ab} c_a^\dagger c_b^\dagger, \quad \eta^- = (\eta^+)^\dagger, \quad \eta^z = \frac{\widehat{N} - L}{2}. \quad (3)$$

These operators generate an  $\mathfrak{su}(2)$  algebra. By unitary congruence,  $J$  can be put into the canonical form  $\bigoplus_{a=1}^L i\sigma_y$ . Thus (3) is the fully multiband, basis-independent version of  $L$  collective pair orbitals;  $J$  may mix bands, spin, orbital labels, momenta, and gauge patches.

For  $0 \leq n \leq L$ , define the normalized AGP state

$$|n\rangle = \left[ \frac{(L-n)!}{n!L!} \right]^{1/2} (\eta^+)^n |0\rangle. \quad (4)$$

The states  $|n\rangle$  form the spin- $S = L/2$  multiplet, with  $m = n - L/2$ .

Let  $A = (A_1, \dots, A_d)$  be an electronic flat connection. Assume that all number sectors are restrictions of one analytic, particle-number-preserving family on the common fermionic Fock space, with the same square factors in every sector:

$$H(A) = \frac{1}{2} \sum_{\lambda=1}^{M_s} S_\lambda(A)^\dagger S_\lambda(A), \quad H_n(A) = H(A)|_{\mathcal{H}_{2n}}. \quad (5)$$

Here  $\mathcal{H}_{2n}$  is the fixed- $2n$ -particle sector, and the  $S_\lambda(A)$  are analytic, Hermitian, particle-number-preserving Fock-space operators. The index  $\lambda$  may include real-space position, QGN channel, and a factorization index of the positive interaction kernel. We impose:

**H1: QGN scalar condition at zero twist.** With  $S_\lambda = S_\lambda(0)$ ,

$$[S_\lambda, \eta^+] = [S_\lambda, \eta^-] = [S_\lambda, \eta^z] = 0. \quad (6)$$

**H2: Frustration freeness.** For the filling sectors considered,

$$S_\lambda |n\rangle = 0 \quad \text{for every } \lambda. \quad (7)$$

**H3: One-body twist source.** The first derivatives

$$B_{\lambda i} = \partial_{A_i} S_\lambda(A) \Big|_{A=0} \quad (8)$$

are number-conserving one-body operators,

$$B_{\lambda i} = d\Gamma(b_{\lambda i}) = \sum_{a,b} (b_{\lambda i})_{ab} c_a^\dagger c_b, \quad (9)$$

with no  $A$ -dependent scalar centering term. Because the square factors are Hermitian, the matrices  $b_{\lambda i}$  are Hermitian. Requiring a Hermitian square-factor representation is part of the restricted theorem.

**H4: Simple analytic branch.** At  $A = 0$ , zero is a simple eigenvalue of  $H_n(0)$  in the fixed- $2n$  sector and is separated from the rest of that finite-dimensional sector by a positive gap. Hence the ground eigenvalue  $E_n(A)$  and eigenvector are analytic near  $A = 0$ . No gap uniform in system size is assumed here.

Define the exact electronic-twist Hessian

$$C_{ij}^{(n)} = \partial_{A_i} \partial_{A_j} E_n(A) \Big|_{A=0}. \quad (10)$$

### 3.1 Automatic H1-H3 for projected PSD-Hubbard models

Let  $\mathcal{V}$  be the spin-up flat-band space, let the spin-down space be its time-reversal conjugate, and define projected orbital vectors

$$v_x = P|x\rangle, \quad p_x = |v_x\rangle\langle v_x|, \quad \sum_x p_x = I_{\mathcal{V}}. \quad (10a)$$

Each  $p_x$  is a rank-one positive operator. It need not be an orthogonal projector because the frame vector  $v_x$  need not be normalized.

The projected PSD-Hubbard factors are

$$S_x = d\Gamma_\uparrow(p_x) - d\Gamma_\downarrow(\bar{p}_x). \quad (10b)$$

For every Hermitian one-body matrix  $a = a^\dagger$ ,

$$[d\Gamma_\uparrow(a) - d\Gamma_\downarrow(\bar{a}), \eta^+] = 0. \quad (10c)$$

Thus H1 is automatic; H2 follows from H1 because the factors annihilate the vacuum; and differentiation of a twisted projector keeps the source one body, proving H3. Uniform pairing of the orbital weights is not needed for these algebraic facts.

### 3.2 Connectivity theorem closing H4 in the Hubbard subclass

Define the projected-frame overlap graph with active microscopic labels as vertices and

$$x \sim y \iff \langle v_x, v_y \rangle = \langle x|P|y \rangle \neq 0. \quad (10d)$$

**Proposition 1 (H4 from frame connectivity).** For the projected PSD-Hubbard Hamiltonian

$$H_0 = \frac{U}{2} \sum_x [d\Gamma_\uparrow(p_x) - d\Gamma_\downarrow(\bar{p}_x)]^2, \quad U > 0, \quad (10e)$$

the AGP is the unique zero-energy state in every total- $2n$  sector,  $1 \leq n \leq L-1$ , if and only if the overlap graph is connected. Consequently the zero eigenvalue has a positive finite-size sector gap and its continuation under an analytic twist is a simple analytic ground branch.

*Proof.* Any zero-energy state is annihilated by every square factor. Summing those null equations and using  $\sum_x p_x = I_{\mathcal{V}}$  gives  $N_\uparrow = N_\downarrow = n$ . Identify

$$\Lambda^n \mathcal{V} \otimes \overline{\Lambda^n \mathcal{V}} \simeq \text{End}(\Lambda^n \mathcal{V}) \quad (10f)$$

by vectorization. If  $A_x = d\Gamma_n(p_x)$ , then a factor acts as

$$S_x \text{vec}(T) = \text{vec}([A_x, T]), \quad (10g)$$

and hence

$$\langle \text{vec}(T), H_0 \text{vec}(T) \rangle = \frac{U}{2} \sum_x \|[A_x, T]\|_{\text{HS}}^2. \quad (10h)$$

Connectedness implies

$$\text{Lie}_{\mathbb{R}}\{ip_x\} = \mathfrak{u}(\mathcal{V}). \quad (10i)$$

To prove (10i), order the active frame vectors along a spanning tree and let  $W$  be the span generated so far. The first nonzero vector generates  $\mathfrak{u}(\text{span } v)$  because  $ip_v$  is a nonzero multiple of the one-dimensional identity. Suppose inductively that  $\mathfrak{u}(W)$  has been generated and that the next frame vector  $v$  enlarges the span. Its overlap with its parent in the tree implies that, in the orthogonal decomposition  $v = w + z$  with  $w \in W$  and  $z \in W^\perp$ , both  $w$  and  $z$  are nonzero. Put

$$p_w = |w\rangle\langle w|, \quad p_z = |z\rangle\langle z|, \quad C_- = |w\rangle\langle z| - |z\rangle\langle w|, \quad C_+ = |w\rangle\langle z| + |z\rangle\langle w|,$$

and  $a = \|w\|^2$ ,  $b = \|z\|^2$ . Since  $ip_w \in \mathfrak{u}(W)$  and  $ip_v$  is one of the generators,

$$\begin{aligned}
[ip_w, ip_v] &= -aC_-, \\
[ip_w, C_-] &= iaC_+, \\
[C_-, iC_+] &= 2i(bp_w - ap_z).
\end{aligned}$$

Thus  $C_-$ ,  $iC_+$ , and  $ip_z$  are generated. Commuting the two cross couplings with  $\mathfrak{u}(W)$  produces  $|u\rangle\langle z| - |z\rangle\langle u|$  and  $i(|u\rangle\langle z| + |z\rangle\langle u|)$  for every  $u \in W$ . Together with  $\mathfrak{u}(W)$  and  $ip_z$ , these span  $\mathfrak{u}(W \oplus \mathbb{C}z)$ . Vectors already lying in  $W$  require no enlargement. Induction reaches all of  $\mathcal{V}$  because the Parseval identity in (10a) implies that the frame spans  $\mathcal{V}$ .

The exterior-power representation of  $U(\mathcal{V})$  on  $\Lambda^n \mathcal{V}$  is irreducible for  $0 < n < L$ . Schur's lemma therefore reduces the commutant in (10h) to scalar multiples of the identity. Its vectorization is the AGP state. Conversely, if the graph has components of span dimensions  $L_1, \dots, L_c$ , the zero-mode dimension is

$$[z^n] \prod_{r=1}^c (1 + z + \dots + z^{L_r}), \quad (10j)$$

which exceeds one at every nonsingular filling when  $c > 1$ . Finite-dimensional positivity then supplies a strictly positive next eigenvalue, and analytic perturbation theory supplies the local analytic branch.  $\square$

For a positive interaction kernel  $H_K = \frac{1}{2} \sum_{x,y} K_{xy} S_x S_y$ , the same result holds whenever  $K$  is strictly positive on the span of the active density-difference constraints. If the kernel loses rank, the correct replacement is irreducibility of the one-body Lie algebra generated by the surviving square factors.

For Model II, the all-size connectivity claim can be proved directly. Define the one-dimensional finite-grid coefficient

$$c_r^{(N)}(\xi) = \frac{1}{N} \sum_{m=0}^{N-1} e^{i\xi \cos k_m} e^{-ir k_m}, \quad k_m = \frac{2\pi m}{N}. \quad (C1)$$

For  $0 < \xi < \pi/2$ ,

$$\text{Re } c_0^{(N)}(\xi) = \frac{1}{N} \sum_m \cos(\xi \cos k_m) > 0, \quad (C2)$$

because every summand is strictly positive. Inversion symmetry of the grid cancels the term odd in  $\sin k_m$  and gives

$$c_1^{(N)}(\xi) = \frac{1}{N} \sum_m \cos k_m e^{i\xi \cos k_m}, \quad \text{Im } c_1^{(N)}(\xi) = \frac{1}{N} \sum_m \cos k_m \sin(\xi \cos k_m) > 0. \quad (C3)$$

The last inequality follows because  $x \sin(\xi x) > 0$  for every nonzero  $x \in [-1, 1]$ , and the grid contains  $k = 0$ . Hence  $c_0^{(N)}$  and  $c_1^{(N)}$  are nonzero for every  $N \geq 2$ . The two-dimensional off-diagonal projector coefficient factorizes, up to a fixed nonzero phase, as

$$\widehat{P}_{12}(r_x, r_y) \propto c_{r_x}^{(N_x)}(\xi) c_{r_y}^{(N_y)}(\xi). \quad (C4)$$

It is therefore nonzero at displacements  $(0, 0)$ ,  $(1, 0)$ , and  $(0, 1)$ . The first edge joins the two orbital layers within a cell, while the latter two join neighboring cells through opposite layers; two-step paths connect equal-orbital vertices across both lattice directions. Thus the projected-frame graph is connected on every rectangular torus for  $0 < \xi < \pi/2$ . The restriction is imposed by the  $c_0$  positivity argument, not by pair-mass positivity.

The analogous all-size Model-I statement is not used below. A separate gauge-invariant audit found connected graphs in the tested Model-I tori and in additional Model-II cases through  $\xi = 2$ , but those results are retained as finite certificates rather than promoted to all-size theorems.

## 4. Frustration-free least-squares curvature

**Lemma 1 (least-squares curvature).** Let  $D(A) : \mathcal{H} \rightarrow \mathcal{K}$  be analytic between finite-dimensional Hilbert spaces and let

$$H(A) = \frac{1}{2} D(A)^\dagger D(A). \quad (11)$$

Suppose  $D(0)|\psi\rangle = 0$  and zero is a simple gapped eigenvalue of  $H(0)$ . Put  $D = D(0)$  and

$$b_i = (\partial_{A_i} D)(0)|\psi\rangle. \quad (12)$$

Then

$$C_{ij} = \text{Re}\langle \Pi b_i, \Pi b_j \rangle, \quad \Pi = P_{\ker D^\dagger}, \quad (13)$$

or, directionally,

$$v_i C_{ij} v_j = \min_{\chi \perp \psi} \|D\chi + v_i b_i\|^2. \quad (14)$$

*Proof.* Choose the analytic gauge  $\langle \psi | \partial_{A_i} \psi \rangle = 0$  and write  $|\chi_i\rangle = \partial_{A_i} |\psi(A)\rangle|_0$ . Since  $D|\psi\rangle = 0$ , the first-order eigenvalue equation is the normal equation

$$D^\dagger(D|\chi_i\rangle + b_i) = 0. \quad (15)$$

The gap makes  $H|_{\psi^\perp}$  invertible, so the normal equation has a unique solution  $|\chi_i\rangle \perp |\psi\rangle$ . Equivalently,  $D|\chi_i\rangle + b_i$  is the orthogonal projection of  $b_i$  onto  $\ker D^\dagger$ . Direct differentiation of  $H = D^\dagger D/2$  gives

$$C_{ij} = \text{Re}\langle D\chi_i + b_i, D\chi_j + b_j \rangle. \quad (16)$$

Equation (13) follows, and (14) is the associated normal-equation characterization. Terms containing  $\partial_i \partial_j D$  vanish because  $D|\psi\rangle = 0$ .  $\square$

For (5), take

$$D_n |\chi\rangle = (S_1 |\chi\rangle, \dots, S_{M_s} |\chi\rangle), \quad b_i^{(n)} = (B_{1i} |n\rangle, \dots, B_{M_s i} |n\rangle). \quad (17)$$

Then Lemma 1 is an exact version of the diamagnetic-minus-paramagnetic Kohn formula, reorganized as a positive least-squares residual.

## 5. Multiband $S \oplus (S - 1)$ source decomposition

The following lemma closes the multiband step. It uses only the skew-unitarity of  $J$  and therefore does not require a band-diagonal, spin-diagonal, or topologically trivial representation.

**Lemma 2 (one-body action on an AGP multiplet).** Assume  $L \geq 2$  and  $1 \leq n \leq L - 1$ . Let  $B = d\Gamma(b)$  be any number-conserving one-body operator and define

$$\beta_B = \langle 1 | B | 1 \rangle = \frac{1}{L} \text{Tr}_b b. \quad (18)$$

There is a lowest-weight pair state  $|\zeta_B\rangle$ , orthogonal to  $|1\rangle$ , in total pseudospin  $S - 1 = L/2 - 1$ , such that



$$B|n\rangle = n\beta_B|n\rangle + \sqrt{\rho_n} U_n(B|1\rangle - \beta_B|1\rangle), \quad (19)$$

where

$$\rho_n = \frac{n(L-n)}{L-1}, \quad (20)$$

and  $U_n$  is the normalized pseudospin-ladder isometry from the one-pair weight space to the  $n$ -pair weight space inside every copy of the  $S-1$  representation.

*Proof.* Since  $B|0\rangle = 0$ ,

$$B(\eta^+)^n|0\rangle = n(\eta^+)^{n-1}[B, \eta^+]|0\rangle. \quad (21)$$

Decompose the two-particle state

$$[B, \eta^+]|0\rangle = \beta_B \eta^+|0\rangle + |\zeta_B\rangle. \quad (22)$$

The coefficient is (18), because the one-particle reduced density matrix of  $|1\rangle$  is  $I_{2L}/L$ . The orthogonal complement of  $\eta^+|0\rangle$  in the two-particle sector is a direct sum of lowest-weight copies of pseudospin  $S-1$ . No lower total pseudospin can occur after acting once with a fermion bilinear on the maximal-spin vacuum.

The AGP norm is

$$\|(\eta^+)^n|0\rangle\|^2 = \frac{n!L!}{(L-n)!}. \quad (23)$$

For a lowest-weight  $S-1$  state,

$$\|(\eta^+)^{n-1}|\zeta_B\rangle\|^2 = \frac{(n-1)!(L-2)!}{(L-n-1)!} \|\zeta_B\|^2. \quad (24)$$

Combining (21)-(24) with the normalization in (4) gives exactly the squared ratio  $\rho_n$  in (20). Because the norm factor in (24) is the same for every lowest-weight vector, the normalized ladder operation defines one linear isometry  $U_n$  on the complete  $S-1$  lowest-weight multiplicity space, not a vector-dependent choice.  $\square$

The identity (18) is the point at which the full multiband pairing matrix simplifies: equal singular values make the one-pair density matrix proportional to the identity on the entire paired flat-band space. If the pairing form has several inequivalent singular-value blocks, the density matrix is not proportional to a single identity and the argument becomes blockwise rather than scalar.

## 6. Exact finite-size reduction identity

For every factor and twist direction, define

$$\beta_{\lambda i} = \langle 1|B_{\lambda i}|1\rangle = \frac{1}{L} \text{Tr } b_{\lambda i}, \quad (25)$$

and the longitudinal tensor

$$\Gamma_{ij} = \text{Re} \sum_{\lambda=1}^{M_g} \beta_{\lambda i}^* \beta_{\lambda j}. \quad (26)$$

It is positive semidefinite.

**Theorem 1 (finite-size restricted QGN reduction).** Under H1-H4, for every  $1 \leq n \leq L - 1$ ,

$$\boxed{C_{ij}^{(n)} = \rho_n C_{ij}^{(1)} + (n^2 - \rho_n) \Gamma_{ij}, \quad \rho_n = \frac{n(L-n)}{L-1}.} \quad (27)$$

In particular,

$$C^{(n)} - \rho_n C^{(1)} \succeq 0. \quad (28)$$

*Proof.* Let  $D = \bigoplus_{n=0}^L D_n$  act on the full even-particle Fock space, with the target carrying the diagonal pseudospin action. Because every  $S_\lambda$  commutes with the pseudospin algebra,  $D$  is an  $SU(2)$  intertwiner. Decompose the target into isotypic components  $\bigoplus_S (V_S \otimes M_S)$ . On each component,  $\ker D^\dagger$  is  $V_S \otimes K_S$  for a multiplicity subspace  $K_S \subseteq M_S$ , and its orthogonal projector has the form  $I_{V_S} \otimes P_{K_S}$ . The normalized ladder maps act as the standard weight-space ladder on  $V_S$  tensored with  $I_{M_S}$ . Therefore the kernel projector commutes with those ladder maps, and distinct total-pseudospin sectors remain orthogonal after projection.

Lemma 2 applied componentwise gives the direct-sum source decomposition

$$b_i^{(n)} = n|n\rangle \otimes \vec{\beta}_i + \sqrt{\rho_n} U_n^\oplus b_{i,\perp}^{(1)}, \quad (29)$$

where  $\vec{\beta}_i = (\beta_{1i}, \dots, \beta_{M_S i})$ . The two terms lie in total pseudospin  $S$  and  $S - 1$ , respectively.

The longitudinal term cannot be removed by the least-squares correction. Indeed,

$$D_n^\dagger(|n\rangle \otimes \vec{v}) = \sum_\lambda v_\lambda S_\lambda^\dagger |n\rangle = 0, \quad (30)$$

where Hermiticity and  $S_\lambda |n\rangle = 0$  were used. Hence  $\Pi_n$  acts as the identity on the longitudinal source. On the  $S - 1$  component, intertwining gives

$$\Pi_n U_n^\oplus = U_n^\oplus \Pi_1. \quad (31)$$

The two pseudospin sectors are orthogonal. Lemma 1 therefore yields

$$C_{ij}^{(n)} = n^2 \Gamma_{ij} + \rho_n (C_{ij}^{(1)} - \Gamma_{ij}), \quad (32)$$

which is (27).  $\square$

**Interpretation.** The  $S - 1$  response channel is the genuine pair-mobility channel. Its reduced matrix element is fixed in the one-pair sector and acquires the hard-core factor  $\rho_n$ . The only possible violation of the exact finite-size filling law in this class is the maximal-pseudospin trace component. It is nonnegative and kinematic, not a new finite-density dressing channel.

## 7. Translation-invariant positive kernels

The factorized notation (5) is convenient for the proof, but locality is clearer before factorizing the interaction kernel. Let  $\Lambda_{\mathbf{N}}$  be a  $d$ -dimensional rectangular torus with side lengths  $\mathbf{N} = (N_1, \dots, N_d)$  and volume

$$V = \prod_{j=1}^d N_j. \quad (33)$$

Suppose there are finitely many translated Hermitian QGN null operators  $X_{R,a}(A)$  and an  $A$ -independent real translation kernel  $\mathcal{V}_{ab}(R - R')$  satisfying

$$\mathcal{V}_{ab}(R) = \mathcal{V}_{ba}(-R) \in \mathbb{R}.$$

Regarded as a matrix on the combined real index  $(R, a)$ , assume  $\mathcal{V}$  is positive semidefinite. It therefore admits a real square root. With  $S_\lambda(A) = \sum_{R,a} W_{\lambda;R,a} X_{R,a}(A)$ , equation (34) equals  $\frac{1}{2} \sum_\lambda S_\lambda(A)^2$ , and every factor is Hermitian, as required in Theorem 1. Explicitly, the Hamiltonian is

$$H(A) = \frac{1}{2} \sum_{R,R' \in \Lambda_{\mathbf{N}}} \sum_{a,b} X_{R,a}(A)^\dagger \mathcal{V}_{ab}(R - R') X_{R',b}(A). \quad (34)$$

We use the finite-torus Fourier convention

$$\widehat{\mathcal{V}}_{ab}(q) = \sum_{R \in \Lambda_{\mathbf{N}}} \mathcal{V}_{ab}(R) e^{-iq \cdot R}, \quad \mathcal{V}_{ab}(R) = \frac{1}{V} \sum_q \widehat{\mathcal{V}}_{ab}(q) e^{iq \cdot R}. \quad (34a)$$

With this convention, positivity is equivalent to  $\widehat{\mathcal{V}}(q) \succeq 0$  for every finite-torus momentum.

Translation invariance of  $|1\rangle$  makes

$$\beta_{a,i}^{(\mathbf{N})} = \langle 1 | \partial_{A_i} X_{0,a}(A) | 0 \rangle | 1 \rangle \quad (35)$$

independent of the cell. The longitudinal tensor is then

$$\boxed{\Gamma_{ij}^{(\mathbf{N})} = V \operatorname{Re} \left[ (\vec{\beta}_i^{(\mathbf{N})})^\dagger \widehat{\mathcal{V}}(0) \vec{\beta}_j^{(\mathbf{N})} \right]}. \quad (36)$$

Equation (36) is invariant under the choice of real square root of the positive kernel. The real-symmetric condition above is the precise Hermitian-kernel hypothesis used in this section.

## 8. Winding formula in arbitrary dimension

Let

$$\mathcal{K}_{\mathbf{N}} = \left\{ \left( \frac{2\pi m_1}{N_1}, \dots, \frac{2\pi m_d}{N_d} \right) : 0 \leq m_j < N_j \right\}. \quad (37)$$

Assume the specified flat-connection lift is translation covariant in the following sense: for each local channel, the normalized one-pair trace can be written as a discrete average of a globally defined periodic scalar symbol  $f_a(k)$ :

$$\langle 1 | X_{0,a}(A) | 1 \rangle = \frac{1}{V} \sum_{k \in \mathcal{K}_{\mathbf{N}}} f_a(k + A) + \text{constant}. \quad (38)$$

Here  $f_a$  is a trace over the flat-band fiber, normalized by the number of pair orbitals per cell. It is gauge invariant.

Write the Fourier series

$$f_a(k) = \sum_{R \in \mathbb{Z}^d} \widehat{f}_a(R) e^{ik \cdot R}. \quad (39)$$

**Proposition 2 (multidimensional roots-of-unity filter).** The finite-grid average obeys

$$\frac{1}{V} \sum_{k \in \mathcal{K}_{\mathbf{N}}} f_a(k + A) = \sum_{\ell \in \mathbb{Z}^d} \hat{f}_a(\ell \odot \mathbf{N}) e^{i(\ell \odot \mathbf{N}) \cdot A}, \quad (40)$$

where  $\ell \odot \mathbf{N} = (\ell_1 N_1, \dots, \ell_d N_d)$ . Consequently,

$$\boxed{\beta_{a,i}^{(\mathbf{N})} = \sum_{\ell \neq 0} i \ell_i N_i \hat{f}_a(\ell \odot \mathbf{N})}. \quad (41)$$

*Proof.* Substitute (39) into the left side of (40). The product of the  $d$  discrete roots-of-unity sums is one exactly when every component of  $R$  is divisible by the corresponding  $N_j$ , and zero otherwise. Differentiation gives (41).  $\square$

Thus the longitudinal term is a pure winding observable. It is absent in the Brillouin-zone integral and can survive at finite size only through Fourier coefficients at displacement vectors that wrap around the torus.

## 9. Locality and topological projector bundles

**Theorem 2 (locality bound).** Assume a fixed number of local QGN channels and a uniform bound

$$\|\widehat{\mathcal{V}}(0)\| \leq V_0. \quad (42)$$

Let  $N_{\min} = \min_j N_j$  and  $N_{\max} = \max_j N_j$ .

1. **Strict finite range.** If there is an  $R_0$  independent of  $\mathbf{N}$  such that

$$\hat{f}_a(R) = 0 \quad \text{whenever } |R|_{\infty} > R_0, \quad (43)$$

then for every torus with  $N_{\min} > R_0$ ,

$$\beta_{a,i}^{(\mathbf{N})} = 0, \quad \Gamma_{ij}^{(\mathbf{N})} = 0. \quad (44)$$

2. **Uniform exponential locality.** If

$$|\hat{f}_a(R)| \leq C_0 e^{-\mu |R|_1} \quad (45)$$

with  $C_0, \mu > 0$  independent of  $\mathbf{N}$ , then there are constants  $C_1, C_2$  depending only on  $C_0, d, \mu$ , the channel number, and  $V_0$  such that, for all sufficiently large  $N_{\min}$ ,

$$|\beta_{a,i}^{(\mathbf{N})}| \leq C_1 N_{\max} e^{-\mu N_{\min}}, \quad (46)$$

and

$$\|\Gamma^{(\mathbf{N})}\| \leq C_2 V N_{\max}^2 e^{-2\mu N_{\min}}. \quad (47)$$

*Proof.* Part 1 follows immediately from (41): every nonzero  $\ell \odot \mathbf{N}$  has  $|\ell \odot \mathbf{N}|_{\infty} \geq N_{\min}$ .

For Part 2, (41) gives

$$|\beta_{a,i}^{(\mathbf{N})}| \leq C_0 N_i \sum_{\ell \neq 0} |\ell_i| \exp \left[ -\mu \sum_j |\ell_j| N_j \right]. \quad (48)$$

The product geometric series is bounded by a constant times  $e^{-\mu N_{\min}}$  for large  $N_{\min}$ , proving (46). Substituting (46) into (36) gives (47).  $\square$

## 9.1 Why topology does not obstruct the estimate

No global Bloch frame is used in (38)-(47). On a gauge patch, the projected one-body symbol is a bundle endomorphism  $x_a(k)$ . On an overlap it transforms by conjugation,

$$x_a^{(q)}(k) = g_{pq}(k)^{-1} x_a^{(p)}(k) g_{pq}(k). \quad (49)$$

The normalized trace  $f_a(k)$  is therefore patch independent and defines a global periodic scalar even when the flat-band vector bundle is topologically nontrivial. The Fourier estimate applies directly to this scalar. A Chern obstruction to a global orthonormal Wannier frame is not an obstruction to a globally defined projector or to gauge-invariant trace symbols.

The locality hypothesis must nevertheless be imposed on the *projected symbol and its specified displacement lift*. A finite-range microscopic parent does not automatically make the spectrally flattened projector strictly finite range; in generic isolated bands one expects exponential rather than compact support.

## 10. Normalization: electronic twist, pair momentum, stiffness, and pair mass

We now freeze conventions. Set the lattice constants, electron charge magnitude, and  $\hbar$  to one.

1.  $A$  is a uniform electronic crystal-momentum shift, implemented as  $k \mapsto k + A$  in the projected interaction.
2. The boundary phase is  $\phi_i = N_i A_i$ . Thus

$$\partial_{A_i} \partial_{A_j} = N_i N_j \partial_{\phi_i} \partial_{\phi_j}. \quad (50)$$

3. A pair contains two electrons, so the physical center-of-mass shift associated with the electronic connection is

$$Q = 2A. \quad (51)$$

4. On a finite torus, ordinary total momenta are discrete. The specified displacement lift nevertheless defines a continuous *lifted pair family* near  $Q = 0$  by

$$\begin{aligned} H_{\text{pair}, \mathbf{N}}^{\text{lift}}(Q) &:= H_1(Q/2)|_{K=0}, \\ E_{\text{pair}, \mathbf{N}}^{\text{lift}}(Q) &:= \lambda_{\min} H_{\text{pair}, \mathbf{N}}^{\text{lift}}(Q) \\ &= E_{\text{pair}, \mathbf{N}}^{\text{lift}}(0) + \frac{1}{2} Q_i (m_{\text{pair}, \mathbf{N}}^{-1})_{ij} Q_j + o(|Q|^2). \end{aligned} \quad (52)$$

This continuous family is fixed by the same lift that defines the electronic twist. It should not, for arbitrary continuous  $Q$ , be identified with an ordinary untwisted finite-torus momentum block: the relative-momentum grid is quantized and generally is not preserved by a shift of  $Q/2$ . When that shifted grid does coincide with the original grid, the usual block identification is recovered.

**Lemma 3 (flat connection is lifted pair momentum).** For a translation-covariant projected interaction in which the electronic flat connection is inserted as  $u_\sigma(k) \mapsto u_\sigma(k + A)$ , the simple one-pair ground branch obeys

$$E_1(A) = E_{\text{pair}, \mathbf{N}}^{\text{lift}}(2A). \quad (52a)$$

Consequently,

$$\boxed{C_{ij}^{(1)} = 4(m_{\text{pair}, \mathbf{N}}^{-1})_{ij}.} \quad (53)$$

*Proof.* Translation covariance decomposes  $H_1(A)$  into discrete crystal-momentum-label sectors. At  $A = 0$  the one-pair AGP state is the simple  $K = 0$  ground state, and H4 ensures that its Kato continuation remains in that label sector for sufficiently small  $A$ . Its physical constituent momenta are  $k + A$  and  $-k + A$ , whose sum is  $2A$ . Definition (52) therefore gives (52a), and the chain rule gives (53). No relabeling of a continuously shifted finite momentum grid is required.  $\square$

When a thermodynamic continuous pair dispersion exists,  $m_{\text{pair}}^{-1}$  denotes its Hessian at  $Q = 0$ . At finite volume,  $m_{\text{pair}, \mathbf{N}}^{-1}$  below always refers to the lifted family (52).

### 10.1 Exact Model-II pair dispersion and positive mass

For Model II,

$$H_{\text{II}}(k) = -t[\sigma_x \sin \alpha_k + \sigma_y \cos \alpha_k], \quad \alpha_k = \xi(\cos k_x + \cos k_y). \quad (\text{M1})$$

Choose the smooth lower-band gauge

$$u_{\uparrow}(k) = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{i\alpha_k/2} \\ ie^{-i\alpha_k/2} \end{pmatrix}, \quad u_{\downarrow}(k) = u_{\uparrow}^*(-k). \quad (\text{M2})$$

The two orbital weights are pointwise constant,  $|u_{1\sigma}(k)|^2 = |u_{2\sigma}(k)|^2 = 1/2$ .

**Lemma 4 (two-line Model-II pair dispersion).** In the one-up/one-down sector on a rectangular torus, the lowest eigenvalue of the lifted pair family (52) is

$$\boxed{E_{\text{pair}, \mathbf{N}}^{\text{lift}}(Q) = \frac{|U|}{4}[1 - |F_{\mathbf{N}}(Q)|], \quad F_{\mathbf{N}}(Q) = \frac{1}{V} \sum_{k \in \mathcal{K}_{\mathbf{N}}} e^{i[\alpha_{k+Q/2} - \alpha_{k-Q/2}].} \quad (\text{M3})$$

For  $N_x, N_y \geq 3$  its exact Hessian is

$$\boxed{(m_{\text{pair}, \mathbf{N}}^{-1})_{ij} = \partial_{Q_i} \partial_{Q_j} E_{\text{pair}, \mathbf{N}}^{\text{lift}}(Q) \Big|_{Q=0} = \frac{|U|\xi^2}{8} \delta_{ij} \succ 0 \quad (\xi \neq 0).} \quad (\text{M4})$$

*Proof.* In the fixed relative-label basis  $k \in \mathcal{K}_{\mathbf{N}}$  of the lifted family, the PSD-Hubbard one-pair matrix is

$$H_Q^{\text{lift}} = |U| \left[ \frac{1}{2} I - \frac{1}{V} \sum_{a=1}^2 |g_{a,Q}\rangle \langle g_{a,Q}| \right], \quad (\text{M5})$$

where, with  $\delta\alpha_k(Q) = \alpha_{k+Q/2} - \alpha_{k-Q/2}$ ,

$$g_{1,Q}(k) = \frac{1}{2} e^{-i\delta\alpha_k(Q)/2}, \quad g_{2,Q}(k) = \frac{1}{2} e^{+i\delta\alpha_k(Q)/2}. \quad (\text{M6})$$

The two nonzero eigenvalues of the rank-two kernel in (M5) are those of

$$\frac{1}{4} \begin{pmatrix} 1 & F_{\mathbf{N}}(Q) \\ F_{\mathbf{N}}(Q)^* & 1 \end{pmatrix}, \quad (\text{M7})$$

namely  $(1 \pm |F_{\mathbf{N}}|)/4$ . The corresponding energies are  $|U|(1 \mp |F_{\mathbf{N}}|)/4$ , while every state orthogonal to the rank-two kernel has energy  $|U|/2$ . Since  $0 \leq |F_{\mathbf{N}}| \leq 1$ , the branch in (M3) is the lowest one. Moreover,

$$\delta\alpha_k(Q) = -2\xi \sum_{i=x,y} \sin k_i \sin(Q_i/2). \quad (\text{M8})$$

On every  $N_i \geq 3$  momentum grid,  $V^{-1} \sum_k \sin k_i = 0$  and  $V^{-1} \sum_k \sin k_i \sin k_j = \delta_{ij}/2$ . Hence  $F(0) = 1$ ,  $\partial_i F(0) = 0$ , and  $\partial_i \partial_j F(0) = -\xi^2 \delta_{ij}/2$ . Differentiating  $|F|$  at the nonzero point  $F(0) = 1$  gives (M4).  $\square$

**Remark (the  $2 \times 2$  anomaly).** The hypothesis  $N_i \geq 3$  is substantive. On a two-point cycle  $k_i \in \{0, \pi\}$ , one has  $\sin k_i = 0$  identically, so the corresponding component of the pair-mass Hessian vanishes. On a  $2 \times 2$  torus both components vanish. This exactly explains the zero-curvature anomaly seen in the earliest finite-size Model-II calculations; it is a discrete-grid exception, not a failure of the formula above.

In the thermodynamic limit the form factor factorizes explicitly,

$$F_\infty(Q) = J_0[2\xi \sin(Q_x/2)] J_0[2\xi \sin(Q_y/2)], \quad (\text{M9})$$

so

$$E_{\text{pair}}(Q) = \frac{|U|\xi^2}{16} (Q_x^2 + Q_y^2) + O(|Q|^4), \quad (\text{M10})$$

which independently reproduces (M4).

**Corollary 4.1 (exact absence of the longitudinal correction).** Let  $s_{Ra}(A)$  be the one-body matrix of the local density-difference factor  $S_{Ra}(A)$  and put

$$w_a(A) = \frac{1}{V} \sum_{k \in \mathcal{K}_{\mathbf{N}}} |u_{a\uparrow}(k + A)|^2.$$

Time reversal makes the corresponding down-spin trace equal to  $w_a(-A)$ . Hence, for  $b_{Ra,i} = \partial_{A_i} s_{Ra}(A)|_0$ ,

$$\text{Tr } b_{Ra,i} = \partial_{A_i} [w_a(A) - w_a(-A)] \Big|_{A=0} = 2 \partial_{A_i} w_a(0). \quad (\text{M10a})$$

In Model II the orbital weights are pointwise constant, so  $w_a(A) = 1/2$  for every  $A$ . Therefore  $\text{Tr } b_{Ra,i} = 0$ ,  $\beta_{Ra,i} = V^{-1} \text{Tr } b_{Ra,i} = 0$ , and

$$\boxed{\Gamma_{ij}^{\text{Model II}} = 0} \quad (\text{M11})$$

Equation (M11) is exact on every finite torus: there is no winding, locality, or finite-size correction to estimate.

With these conventions, define the finite-volume canonical Drude curvature

$$\kappa_{ij}^{(n)}(V) = \frac{1}{4V} C_{ij}^{(n)}. \quad (54)$$

Let  $N_{\text{flat}}$  count flat bands including spin. The number of pair orbitals is

$$L = \frac{N_{\text{flat}}}{2} V, \quad \nu = \frac{n}{L} = \frac{2n}{N_{\text{flat}} V}. \quad (55)$$

Substituting (27) and (53) into (54) gives the exact finite-size formula

$$\boxed{\kappa_{ij}^{(n)} = \frac{N_{\text{flat}}}{2} \nu(1-\nu) \frac{L}{L-1} (m_{\text{pair}, \mathbf{N}}^{-1})_{ij} + \frac{n^2 - \rho_n}{4V} \Gamma_{ij}.} \quad (56)$$

When  $\Gamma = 0$ , the only difference from the thermodynamic conjecture is the universal canonical factor  $L/(L-1)$ . For an anisotropic tensor, compare along a fixed real direction  $v$ . Define

$$R_{\mathbf{N},v} = \frac{v_i \kappa_{ij}^{(n)} v_j}{(N_{\text{flat}}/2) \nu (1-\nu) v_i (m_{\text{pair},\mathbf{N}}^{-1})_{ij} v_j}. \quad (57)$$

Using  $C^{(1)} = 4m_{\text{pair},\mathbf{N}}^{-1}$ ,

$$R_{\mathbf{N},v} = \frac{L}{L-1} + \frac{L^2(n-1)}{(L-1)(L-n)} \frac{v_i \Gamma_{ij} v_j}{v_i C_{ij}^{(1)} v_j}. \quad (58)$$

## 11. Thermodynamic restricted reduction theorem

**Theorem 3 (thermodynamic QGN reduction).** Consider a sequence of regular tori with  $N_{\min} \rightarrow \infty$  and bounded aspect ratio. Assume H1-H4 at each size, a fixed finite-range or uniformly exponentially local lift as in Theorem 2, and fillings  $n_{\mathbf{N}}/L \rightarrow \nu \in (0,1)$ . Suppose that in a direction  $v$  the one-pair inverse mass remains nonzero:

$$\liminf_{\mathbf{N} \rightarrow \infty} v_i (m_{\text{pair},\mathbf{N}}^{-1})_{ij} v_j > 0. \quad (59)$$

Then

$$R_{\mathbf{N},v} \rightarrow 1. \quad (60)$$

For fixed strict finite range,  $\Gamma^{(\mathbf{N})} = 0$  exactly once the torus exceeds the support, and

$$R_{\mathbf{N},v} = \frac{L}{L-1} \quad (61)$$

at every larger size. For uniform exponential locality, the second term in (58) is bounded by a polynomial in the side lengths times  $e^{-2\mu N_{\min}}$ , and therefore vanishes.

*Proof.* The finite-range statement follows from (44), (56), and (58). In the exponential case, (47) and  $L = (N_{\text{flat}}/2)V$  show that the longitudinal correction in (58) is bounded by a polynomial factor times  $e^{-2\mu N_{\min}}$ . Assumption (59) prevents division by a vanishing pair curvature. The first term in (58) tends to one.  $\square$

If the lifted finite-volume mass converges to a thermodynamic pair mass,  $(m_{\text{pair},\mathbf{N}}^{-1})_{ij} \rightarrow (m_{\text{pair}}^{-1})_{ij}$ , then (60) is the corresponding thermodynamic stiffness/pair-mass equality.

Gao, Han, and Khalaf define their target response through the second derivative of the ground-state energy under a flat connection and denote the thermodynamic quantity  $D_s$  [1]. With that literal definition, no additional charge-gap hypothesis is needed to state their conjecture. Their finite-size calculations are more precisely Drude curvatures, and a stricter dynamical identification with a zero-frequency transport coefficient may still require an order-of-limits argument.

### 11.1 Headline application: Model II Hubbard

**Theorem 4 (Gao-Han-Khalaf equality for connected Model-II tori).** Consider the lower spin-degenerate flat band of (M1), projected with the PSD-Hubbard interaction

$$H(A) = \frac{|U|}{2} \sum_{R,a} [\bar{n}_{R\uparrow}(A) - \bar{n}_{R\downarrow}(A)]^2, \quad (\text{M12})$$



and the flat connection  $k \mapsto k + A$ . Let  $\xi \neq 0$  and let  $N_x, N_y \geq 3$ . On any rectangular torus for which the projected-frame graph is connected, the fixed-number ground branch is unique and analytic and, for  $1 \leq n \leq V-1$ , its exact curvature is

$$\kappa_{ij}^{(n)}(V) = \frac{1}{4V} C_{ij}^{(n)} = \frac{V}{V-1} \nu(1-\nu) \frac{|U|\xi^2}{8} \delta_{ij}, \quad \nu = \frac{n}{V}. \quad (\text{M13})$$

The mass formula (M4) holds for every  $\xi \neq 0$ ; the restriction  $0 < \xi < \pi/2$  enters only through the all-size connectivity proof. In particular, for  $0 < \xi < \pi/2$  the frame graph is connected on every rectangular torus, and for any regular thermodynamic sequence with  $n_V/V \rightarrow \nu \in (0, 1)$ ,

$$D_{s,ij} := \lim_{V \rightarrow \infty} \kappa_{ij}^{(n_V)}(V) = \nu(1-\nu)(m_{\text{pair}}^{-1})_{ij} = \frac{|U|\xi^2}{8} \nu(1-\nu) \delta_{ij}. \quad (\text{M14})$$

Thus the finite-size theorem applies at any audited-connected value of  $\xi \neq 0$ , while the unconditional all-torus thermodynamic statement is proven throughout  $0 < \xi < \pi/2$ .

*Proof.* Time reversal gives H1-H3. Connectedness of the projected-frame graph gives H4 by Proposition 1. For  $0 < \xi < \pi/2$  that connectedness is proven uniformly for every rectangular torus; outside this interval the same finite-size conclusion holds whenever connectedness is independently certified. Equation (M11) removes the longitudinal term exactly. With  $L = V$  and  $N_{\text{flat}} = 2$ , Theorem 1 and Lemmas 3-4 give

$$C_{ij}^{(n)} = \frac{n(V-n)}{V-1} C_{ij}^{(1)} = \frac{n(V-n)}{V-1} \frac{|U|\xi^2}{2} \delta_{ij}. \quad (\text{M15})$$

Dividing by  $4V$  proves (M13); taking  $V \rightarrow \infty$  proves (M14).  $\square$

## 12. Finite-volume isolation and the deferred sector-gap problem

Proposition 1 gives a unique AGP zero mode and a strictly positive spectral gap on every fixed connected torus. That finite-volume isolation is the only gap input used by Lemma 1 and Theorem 1. Neither Theorem 3 nor Theorem 4 assumes a lower bound uniform in volume.

The representation-theoretic decomposition of the full sector spectrum, comparisons of gaps across fillings, and the rank-one Aldous question are mathematically interesting but are not load-bearing for the stiffness theorem. They are deferred to a standalone mathematical note. No finite-size sector-gap ordering or  $O(N^{-2})$  scaling claim is asserted or used in this paper.

This separation is deliberate: the Gao-Han-Khalaf conjecture addressed by Theorem 4 is stated in terms of thermodynamic flat-connection ground-energy curvature. A stricter dynamical identification and its order of limits are scoped as a subsequent project in Section 13.3.

## 13. Relation to prior work and theorem boundary

### 13.1 What is new relative to earlier flat-band results

Several ingredients of the proof have important precedents.

Tovmasyan *et al.* established exact paired ground states and an emergent pseudospin  $SU(2)$  structure for projected attractive Hubbard models under the uniform-pairing condition [5]. Huhtinen *et al.* derived an exact Cooper-pair mass in that setting and related it to the minimal quantum metric, resolving orbital-embedding dependence [10]. The QGN construction generalized the solvable-ground-state framework and computed few-body excitations and pseudospin-based stiffness information for ideal models [2]. More recently, Herzog-Arbeitman *et al.* analytically classified pair bound-state spectra in exact eta-pairing flat-band models and identified the

minimal metric in the quadratic pair dispersion [11]. Gao, Han, and Khalaf then supplied rigorous many-body lower bounds, the pair-mass upper bound, and the equality conjecture addressed here [1].

A targeted prior-art sweep through July 23, 2026 found no earlier proof of the exact Model-II many-body stiffness/pair-mass identity. The closest results establish exact paired ground states, exact or geometric pair masses, solvable pair spectra, stiffness bounds, and numerical saturation, but not the finite-density equality proved here.

The present contribution is different in four specific respects:

1. it proves an exact fixed-particle-number twist-Hessian reduction, including the finite-size factor  $L/(L-1)$  and the complete nonnegative defect tensor  $\Gamma$ ;
2. it proves that the defect is a winding trace and therefore vanishes exactly for sufficiently large fixed finite-range lifts and asymptotically for uniformly exponentially local lifts;
3. it gives a necessary-and-sufficient connectivity criterion for uniqueness of the AGP zero mode in projected PSD-Hubbard models; and
4. it combines these results with an exact Model-II pair branch to prove the GHK equality, rather than infer it from mean field, a variational upper bound, or numerical saturation.

The connectivity theorem is a superconducting operator-space analogue of the irreducibility criteria in rigorous flat-band ferromagnetism. Mielke proved necessary-and-sufficient uniqueness conditions in terms of irreducibility of the one-particle density matrix [8,9]. Here vectorization converts the paired zero-mode problem into a commutant on  $\Lambda^n \mathcal{V}$ , and connected rank-one projected orbitals generate the full one-body unitary algebra.

### 13.2 Non-Hermitian QGN boundary

Perfect QGN is a property of the flat-band electronic structure; it should not be conflated with the narrower interaction class used in this theorem. The standard ideal Hamiltonians of Ref. [2] are built from Hermitian local operators and a positive kernel. In its “Possible Extensions” section, the same work explicitly introduces an SSH-type pairing construction with a non-Hermitian interaction coefficient/form, describes it as a generalization of the standard form, and notes that its excitations are not exactly solvable [2].

Accordingly:

- a band used in such an extension may still satisfy perfect QGN;
- the associated interaction need not admit the Hermitian positive-square least-squares and pseudospin-intertwiner structure used here;
- the large discrepancy found in our non-Hermitian generalized-nesting stress test is evidence that Hermiticity is a substantive hypothesis, but that random test was not a reproduction of the published SSH extension and is not advertised as a counterexample to it.

The restricted theorem therefore proves the Hubbard Model-II case and the stated Hermitian positive-square class. It does not adjudicate the equality for every non-Hermitian extension that might fall under the broadest phrase “QGN model.”

### 13.3 Remaining scope

The theorem does not presently cover degenerate fixed-number ground spaces, several inequivalent singular-value blocks of the pairing matrix, rank-deficient interaction kernels without a separate Lie-algebra audit, or general non-Hermitian generalized-QGN constructions. The reducible counterexample classifies the first boundary exactly. The natural replacement in the multi-block case is a block-resolved filling law with several pseudospins.

Ref. [1] calls the thermodynamic ground-energy curvature under a flat connection  $D_s$ , and that literal conjecture is what Theorem 4 proves for Model II. A stricter identification with a dynamical zero-frequency superfluid response can depend on the order of thermodynamic, zero-momentum, and zero-frequency limits. No uniform-in-volume gap is assumed or needed here, and no sector-gap scaling statement is used. We leave the dynamical order-of-limits problem as the next project rather than attaching it as an unnecessary hypothesis to the result proved here.

## 14. Reproducibility and cross-package verification

The consolidated archive contains independent certificates for every stage of the argument.

| Component                      | Principal check                                                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Winding obstruction            | Exact SymPy Kohn response; independent fixed-number projected-density implementation; resonant and nonresonant controls.       |
| Reducible obstruction          | Fixed-local range-two ED; exact zero-curvature component-polarized state; positive one-pair mass.                              |
| General reduction              | Random full-rank multiband skew-unitary pairing matrix; exact tensor identity and direct finite-difference cross-check.        |
| Longitudinal formula           | Independent trace-derivative certificate source for the filling-dependent defect.                                              |
| H4 connectivity                | Connected and disconnected random Parseval frames; full Lie dimensions and exact zero-mode counts; 55 Model-I/II graph audits. |
| Model-II closure               | Direct lifted $Q$ -family versus the closed formula, $Q = 2A$ , mass Hessian, exact $\Gamma = 0$ , and many-body curvature.    |
| Sector-gap evidence            | 160 rank-one frame cases and independent Gram-versus-dense validations; retained as non-load-bearing evidence.                 |
| Published-model implementation | Independent projected-Hubbard ED code for Gao-Han-Khalaf Models I and II is included in the archive.                           |

The supplied consolidated output files report PASS throughout. The exact four-cell winding certificate independently gives the raw defect  $2/3$  and the normalized defect  $1/24$  at  $g = 1$ , matching (O5). The general reduction certificate has maximum tensor-identity error  $1.25 \times 10^{-18}$  and maximum finite-difference error  $5.17 \times 10^{-8}$ . The H4 certificate attains the full dimensions 16 and 25 for  $u(4)$  and  $u(5)$ , reproduces every connected and disconnected nullity count, finds all 55 audited Gao-Han-Khalaf frame graphs connected, and has maximum Parseval-frame error  $8.10 \times 10^{-16}$ . The Model-II certificate agrees with the direct lifted pair matrix to  $4.26 \times 10^{-16}$ , verifies the electronic-twist identity to  $5.88 \times 10^{-16}$ , reproduces the finite-grid mass Hessian to  $3.56 \times 10^{-9}$ , and reproduces the many-body finite-size formula to  $2.50 \times 10^{-9}$ . The separate sector-gap certificate finds filling-independent gaps in 160 retained rank-one cases to worst relative deviation  $1.02 \times 10^{-13}$ , while a generic non-rank-one Hermitian control has gap ratio 0.432639; these gap observations are not used in the proof.

The original counterexamples are now boundary tests of the final theorem rather than disconnected side projects. The winding defect is exactly the longitudinal tensor in a nonlocal resonant lift, and the even/odd counterexample is exactly the disconnected ground-space degeneracy counted by the H4 theorem.

## 15. Conclusion

The research arc began with a broad attempt to prove an exact many-body/two-body reduction and initially produced two counterexamples. The first showed that continuous finite-torus twist curvature is not determined by untwisted samples unless the displacement lift is fixed. The second showed that even a fixed-local, twist-stable QGN model can evade the pair-mass value when its paired ground space is reducible. Those failures identified, rather than destroyed, the theorem.

Under H1-H4 in the Hermitian positive-square class, the exact finite-size identity is

$$C_{ij}^{(n)} = \frac{n(L-n)}{L-1} C_{ij}^{(1)} + \left[ n^2 - \frac{n(L-n)}{L-1} \right] \Gamma_{ij}, \quad \Gamma \succeq 0. \quad (62)$$

The longitudinal tensor is a winding observable. Fixed locality removes it. In the projected PSD-Hubbard subclass, projected-frame connectivity is exactly the condition that supplies H4 by removing competing AGP occupancy sectors. In Model II, the cancellation is stronger:  $\Gamma = 0$  exactly on every finite torus, with no correction, while the one-pair mass is strictly positive and known in closed form. The resulting finite-size formula (M13) and thermodynamic equality (M14) prove the restricted Gao-Han-Khalaf conjecture for the projected-Hubbard Model II throughout  $0 < \xi < \pi/2$ , and on any additional finite torus whose frame connectivity is certified.

The next physics project is extension, not completion of this case: determine the largest Hermitian QGN interaction class for which the intertwiner proof survives, derive the multi-singular-value block law, and address the dynamical order of limits under transport definitions stricter than the flat-connection curvature used in Ref. [1].

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